



Tribhuvan University
Faculty of Humanities and Social Sciences

“CRIME REPORTING SYSTEM ”

A PROJECT REPORT

Submitted to
Department of Computer Application
Morgan International College

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by
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Tribhuvan University
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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project is prepared under my supervision by ATIT SHARMA entitled **“CRIME REPORTING SYSTEM”** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application in recommendation for the final evaluation.

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LETTER OF APPROVAL

This is to certify that this project prepared by ATIT SHARMA entitled “**CRIME REPORTING SYSTEM**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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ABSTRACT

This platform will not be limited to reporting only but it also helps in investigating reported cases. Online reporting is a mechanism that helps to report any kind of criminal case through the internet. Criminal cases never end therefore this system will be very useful to easily report on criminal cases.

Keywords: Html, Css, Python, PostgreSQL, Django

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LIST OF ABBREVIATIONS

CSS	Cascading Style Sheets
DB	Database
HTML	HyperText Markup Language
IDE	Integrated Development Environment
JS	JavaScript
CRS	Crime Reporting System
OTP	One time password
SQL	Structured Query Language
UI	User Interface
UX	User Experience

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CHAPTER: 1

INTRODUCTION

1.1 Introduction

Crime continues to rise globally, and Nepal ranks 76th in terms of crime rate. Many people still struggle to access justice. To help address this issue, we are developing an online system that allows users to report criminal cases through the internet. This platform goes beyond reporting - it also supports the investigation of cases, making the process faster and more accessible.

The Crime Reporting System is a web-based application designed to be user-friendly. Anyone with internet access and basic knowledge of how to use a website can file a complaint without needing to visit a police station. This saves time for victims and streamlines the reporting process.

The system includes three main interfaces. The public portal is where users can file complaints. The admin portal manages and verifies reports, ensuring they are genuine. Finally, the police portal is operated by the police office to handle and take action on verified reports. This system is an effective solution for reporting misconduct online in a more convenient and organized way.

1.2 Problem Statement

After an extensive analysis of the system, we have set some goals and made some analysis on the problem that can arise during and after the completion of the system. We have researched by interviewing with different people to gather about difficulties about criminal cases and we found these problems in existing system:

- To Existing system is based on hard copy system which is time consuming, costly in terms of time, money and resources.
- Existing system is not available for anywhere and any time.
- Sometimes police refuse to accept report from the victim.

1.3 Objective

The objective of this is to develop a crime reporting system. Expected achievements in order to fulfill the objective are:

- To eliminate the paper base system, reduce time, cost, money and resources.

- To make police available 24 hours in 7 days.
- To develop a secure and platform for crime reporting.

1.4 Scope and Limitation

1.4.1 Scope

The scope of the online crime reporting system includes efficient management of crime reports between users and authorities through a centralized digital platform. The system enables users to register, file complaints, and monitor the progress of their reports in real time. Police officials can manage and update the status of cases, while administrators oversee the entire platform, ensuring smooth operation and coordination. The database maintains detailed records of crime reports, user profiles, categories of crimes, and report history. It also supports status tracking through remarks such as pending, under investigation, closed, or rejected, which enhances transparency throughout the process. The system ensures secure login and timestamp tracking for all critical actions. Additionally, features like system backup and restore help maintain data integrity, while access control ensures that only authorized users can view or modify sensitive information. Overall, the platform improves the traditional reporting process by allowing online complaints, real-time updates, and centralized monitoring for all involved parties.

1.4.2 Limitation

- User interface is only in English.
- Internet reliance means internet is not everywhere yet so that area cannot be connected.
- People who are not familiar with computers can't use this application.

1.5 Development Methodology

For the development of this system, Object oriented approach is used that includes object and class diagram, state and sequence diagram, activity diagram, component diagram.

1.6 Report Organization

Chapter 1 deals with the introduction of the system with objective scope and limitation.

Chapter 2 summarizes the work that has been carried out in the field of crime reporting and also describe the feature about existing system related to Crime Reporting System.

Chapter 3 focuses on the different requirements of the system which describe the functional, non-functional feasibility study, Object Modelling: Object and class Diagram,

Dynamic Modelling: State and Sequence Diagram, Process Modelling: Activity Diagram design of the system with Component Diagram, and Deployment Diagram. Implementation of Algorithm with its details.

Chapter 4 emphasizes tools used in system development, implementing details and result of test performed.

Chapter 5 highlights brief summary of the lesson learnt, outcome and conclusion of the whole system and explain what have been done and what further improvements could be done.

CHAPTER: 2

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

As we know that internet access is everywhere, People are connected with the internet throughout the world. That is why we have developed this web based Crime Reporting System so that any kind of criminal case can be reported with the help of the internet. There are many benefits of this system over the existing system. It makes it an easy platform to report crime. People do not need to travel to a police station to submit an FIR and it also helps to save the victim. It also avoids the paper based reporting system. For police station this system helps to eliminate the inconvenience of the file system to record the report. Using database management system all the records can be kept within a small place and data can be saved for long duration. Additional manpower reduces if we use this web based crime reporting system. The existing system is traditional based which has many disadvantages. In remote area people are unable to get justice because of lack of police station access. That creates inconvenience to the victim. Sometimes police refuse to receive files, in such conditions the traditional physical crime reporting system failed to give justice to the victim. So our system has tried to give a suitable and more user friendly platform to file crime.

2.2 Literature Review

The Crime Reporting System is a digital platform that allows citizens to report crimes directly to law enforcement through the internet, eliminating the need for physical presence at police stations. This system provides convenience, saves time, and ensures timely action by enabling real-time case tracking and communication between the complainant and authorities. As described in the Crime Reporting System Overview [1], such platforms enhance efficiency by offering functionalities like user registration, complaint filing, and status monitoring. They also help reduce paperwork and minimize delays often experienced in traditional reporting methods.

To further simplify the First Information Report (FIR) process, the Online FIR Registration and Crime Reporting System [2] was introduced. This system allows users to submit FIRs online and track their progress through a secure portal. It helps bridge the gap between the public and the police, especially in rural or remote areas where accessing police stations

may be difficult. The platform supports categorizing crime types, attaching evidence, and maintaining a digital record of complaints, which helps improve transparency and follow-up.

According to GeeksforGeeks [3], online crime reporting systems come with numerous advantages. They offer 24/7 accessibility, reduce human interaction for sensitive cases, and maintain digital trails for accountability. These systems are particularly beneficial for victims hesitant to report incidents in person due to fear or stigma. Additionally, digital reports can be easily managed, filtered, and analyzed to understand crime trends and improve policing strategies.

The development process for such systems can follow the Prototype Model, as outlined by S. Kumar [4]. This model allows for early user feedback by developing functional prototypes, making it ideal for public-facing applications like crime reporting platforms. It ensures the system is tailored to real user needs, allows for incremental improvements, and reduces the risk of missing critical features during the initial design.

CHAPTER: 3

SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

- This system is designed with the series of processes starting with requirement analysis, design, implementation, testing and maintenance. During requirement analysis, all the functional and non-functional requirement are analyzed and system is developed according to the requirement then designing of the system is carried out. After the design process, coding and development part is started then after integrating the system there is testing of the system. If the testing is positive then system is implemented otherwise some maintenance is done and system come in operation.

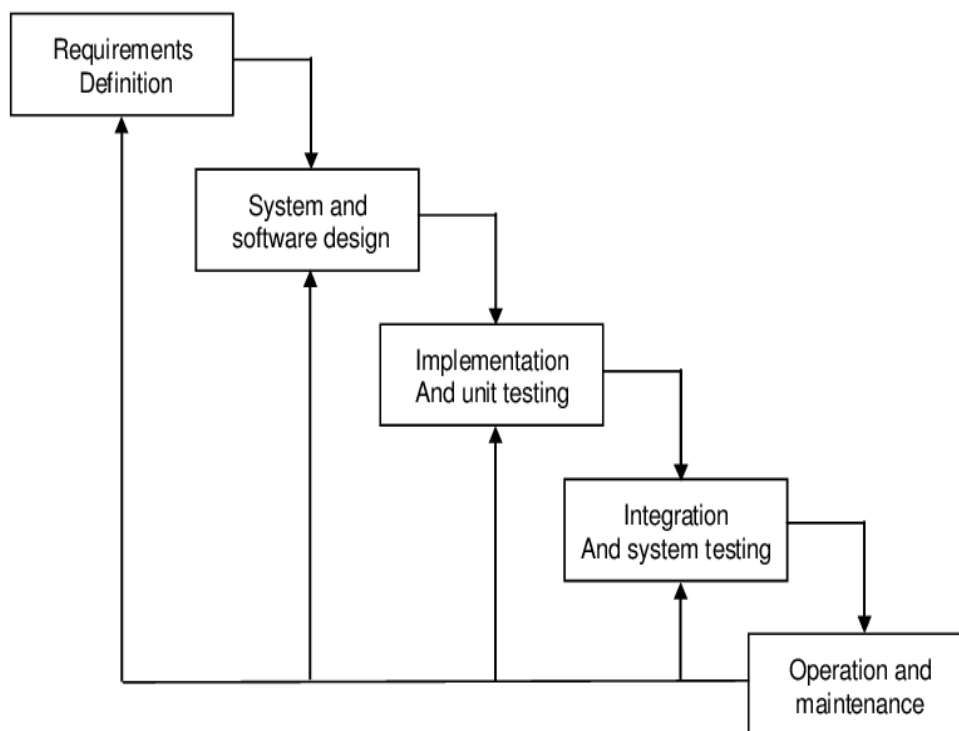


Figure 1 Waterfall Methodology for Crime Reporting System

Why Waterfall Model?

- **Clear Requirements** - Since the project had well-defined objectives (online crime reporting, police registration, real-time updates), the linear progression of Waterfall ensured no ambiguity in development.

- **Documentation-Driven** - Each phase (requirements, design, implementation, testing, deployment) had detailed documentation, making it easier for stakeholders (police officers, developers) to track progress.
- **Budget & Time Control** - Unlike Agile (which allows frequent changes), Waterfall's rigid structure helped in sticking to the initial budget and timeline.
- **Suitable for Small-Scale Projects** -As this system had a fixed scope (no frequent feature additions), Waterfall was more efficient than iterative model.

Phases of Waterfall Model in This Project

• Requirement Gathering & Analysis

Conducted surveys with police stations and users in Nepal to identify needs (e.g., Online FIR system, real time status check).

Defined functional (status changing) and non-functional (security, scalability) requirements.

• System Design

Created ER diagrams for database structure (PostgreSQL).

Designed UI/UX mockups (HTML, CSS, Bootstrap) for a mobile-friendly interface.

• Implementation (Coding)

Frontend: Developed using HTML, CSS, JavaScript.

Backend: Built with Python for business logic.

• Testing

Unit Testing – Checked individual modules (login, registrations, file case).

Integration Testing – Verified if frontend, backend, and DB worked seamlessly.

• Deployment & Maintenance

Hosted on a cloud server (AWS/Google Cloud) for scalability.

Provided training to police station staff for smooth adoption.

3.1.1 Requirement Analysis

The requirements are to be collected before starting project's development life cycle. To design and develop system, functional and non-functional requirements of the system has been studied.

i. Functional Requirements User Management

- Users can and check status of their complaints.
- Admins and police users can manage report details and change details.

File a Complaint Report

- Police and Admin can list complaints items with descriptions, images, and titles.
- Ability to update, add, or remove complaints items.

Report Management

- Users can register, check statys, and delete crime report.
- Police receive and process reports.

USECASE DIAGRAM

The use case diagram of a crime reporting system illustrates the interaction between various users and the system's functionalities. The primary actors involved are the users, police, admin. Users can anonymously report a crime, view the report . The admin manages user accounts, updates the crime report , monitors complaints. Police staff are responsible for viewing incoming complaints, updating the status of complaints(such as pending, investigating, or finished), and handling complaint coordination. This diagram helps visualize how each actor uses the system, outlining the overall scope and main operations of the application.

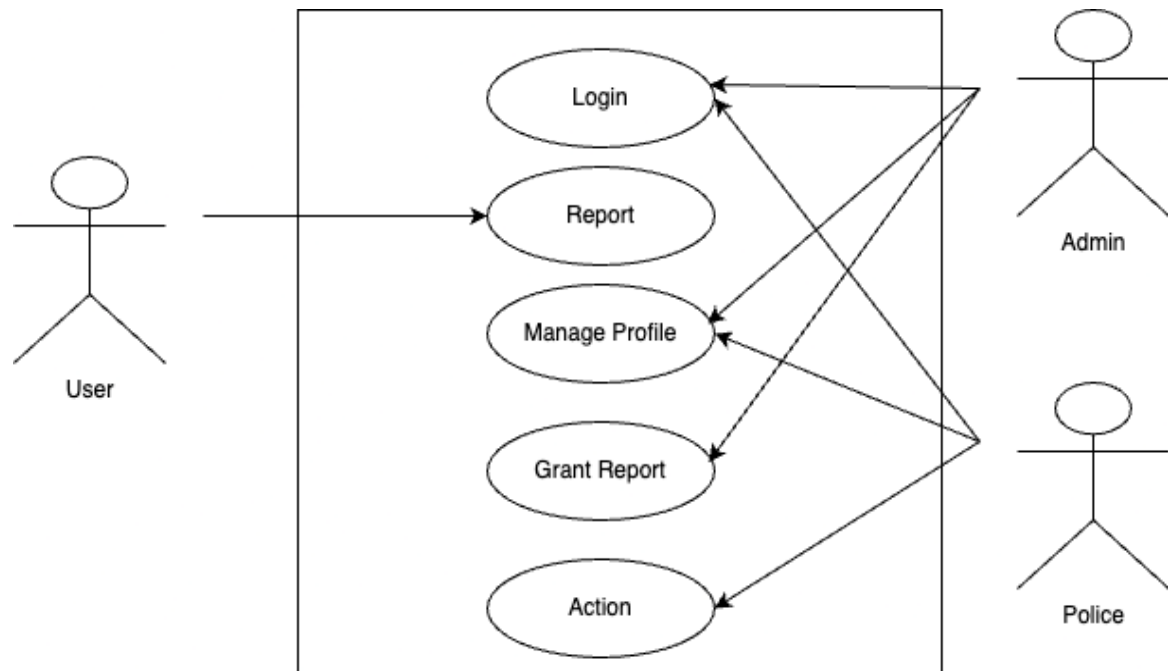


Figure 2 Use case Diagram of CRS

ii. Non-Functional Requirements

- Performance – The system should process reports and updates within seconds.

- Scalability – Must handle high user traffic and multiple complaints listings.
- Security – Secure login, password encryption, and user data protection.
- Availability – 24/7 uptime to ensure continuous service.
- Usability – User-friendly UI with simple navigation.
- Compatibility – Should work across web and mobile platforms (Android & iOS).

3.1.2 Feasibility Analysis

A feasibility study evaluates a project's potential for success by examining key factors including technical, operational, and economic considerations.

• Technical Feasibility

The system is designed to efficiently handle high volumes of simultaneous complaints while ensuring accuracy and reliability. Since it uses open-source technologies (such as Python/PostgreSQL), development and maintenance costs remain low. The required hardware and software are budgetfriendly and widely available, making implementation straightforward. Additionally, the real-time database ensures instant updates on complaints statuses, case records, and user data. With no major technical constraints, this system is fully feasible for deployment.

• Operational Feasibility

The software is user-friendly and requires minimal training, eliminating the need for specialized staff. It streamlines police stations operations by automating complaints management, reducing manual errors, and improving efficiency. Users can access all necessary information (title, status, other details) without direct police interaction. The system also includes validation checks to prevent data entry mistakes, ensuring smooth functionality. Since it aligns with organizational needs, police stations can seamlessly integrate it into daily operations.

• Economic Feasibility

The cost of development and deployment is minimal, as the system relies on affordable, open-source solutions. After implementation, it reduces operational expenses by cutting down on manual labor and minimizing reporting errors. The longterm benefits—such as faster service, improved users satisfaction, and increased complaints accuracy—outweigh the initial investment.

• Schedule Feasibility Study

Table 1 Gantt chart Table for Crime Reporting System

Task Name	Duration
Getting Started	2 weeks
System Design & Architecture	2 weeks
Implementation	7 weeks
Deployment	5 weeks
Documentation	12 weeks

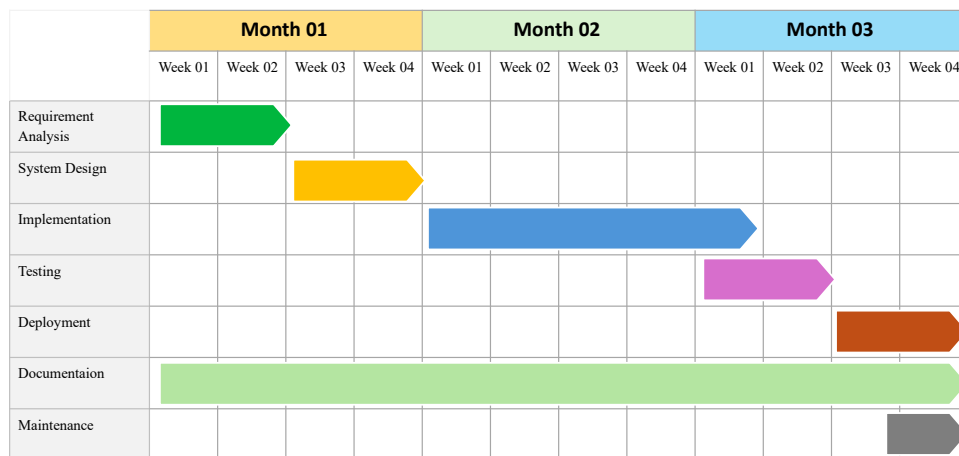


Figure 3 Gantt Chart for Crime Reporting System

3.1.3 Object Modelling: Class Diagram

The class diagram of the Crime Reporting System illustrates the system's core structure by depicting its primary classes and their relationships. It includes classes such as Admin, User, Police, User Report. The Admin oversees system operations, while Police can register, log in, and see Complaints lists.. This diagram helps visualize the interaction between entities, ensuring a clear and organized development process for the system.

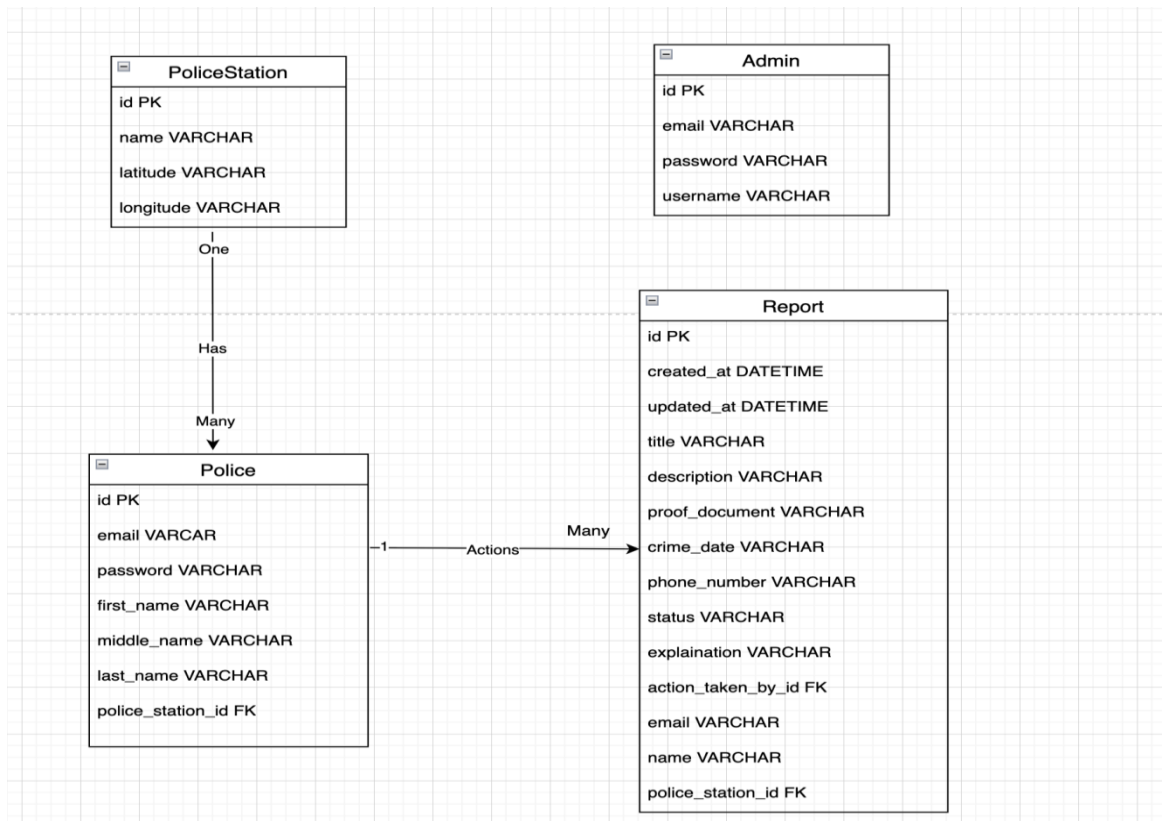


Figure 4 Class Diagram of Crime Reporting System

3.1.4 Dynamic Modelling: State and Sequence Diagram

For Admin and Police

The admin sequence diagram details the steps followed by the administrator to manage the crime report on the system. The admin logs into the backend interface and gains access to various modules including users, police stations, reports. Upon receiving new report, the admin views the report details and updates the status (e.g., pending, investigating, or action taken) based on police processing. Additionally, the admin can add change theme according to police stations. The sequence illustrates the admin's central role in maintaining operational efficiency and ensuring seamless user transactions.

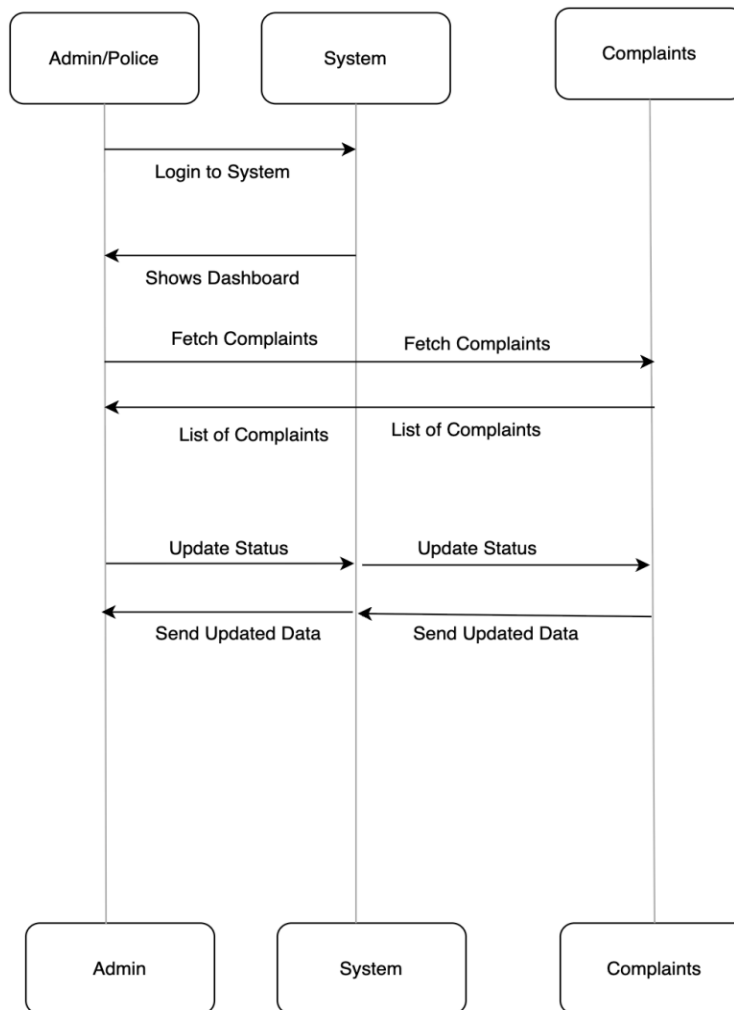


Figure 5 Sequence Diagram for Admin/Police of Crime Reporting System

For User The user sequence diagram outlines how a user interacts with the system to file a complaint. The sequence starts when the user enters into the platform and browses public complaint reports and their status. After selecting file report button, the user redirects to a form, the users fill the form, which is then saved to the database. The system provides complaint confirmation and displays the complaint status. The user can view complaint history, track progress (e.g., in pending, investigating, or action taken), and receive remarks or updates. This diagram highlights the real-time interaction flow from user authentication to complaint management and case status tracking.

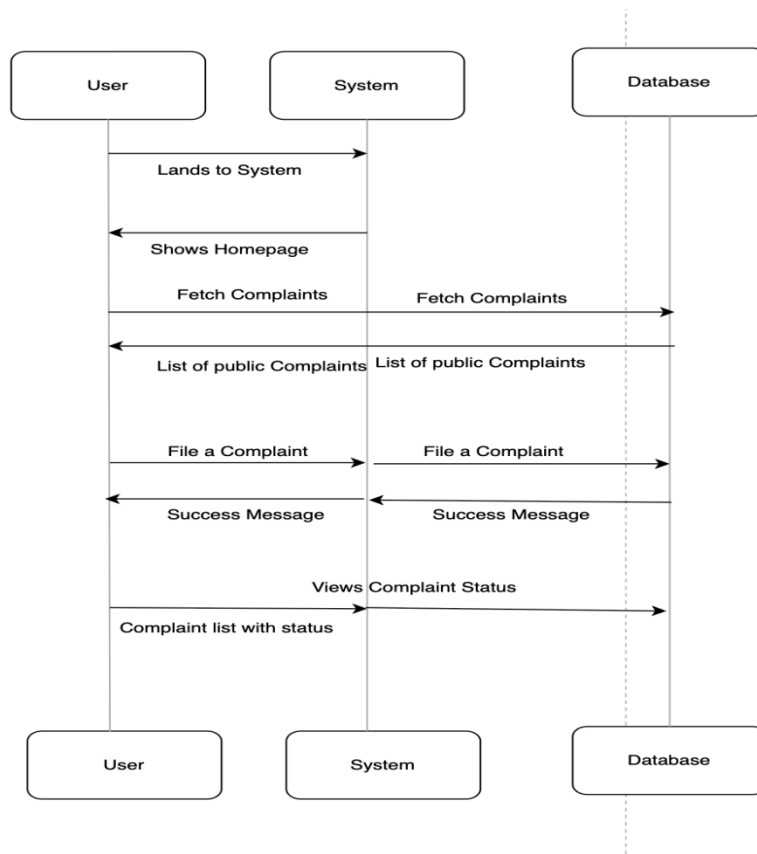


Figure 6 Sequence Diagram for User of Crime Reporting System

3.1.5 Process Modelling: Activity Diagram

The activity diagram for the Crime Reporting System showcases the police and admin interactions within the system. For the user, the process begins with login or registration. After this, the police track complaint status and view complaints. On the admin side, the activity diagram illustrates the admin's workflow, starting with login, managing police stations and police users, adding or updating complaints, themes, and reviewing and updating complaints.

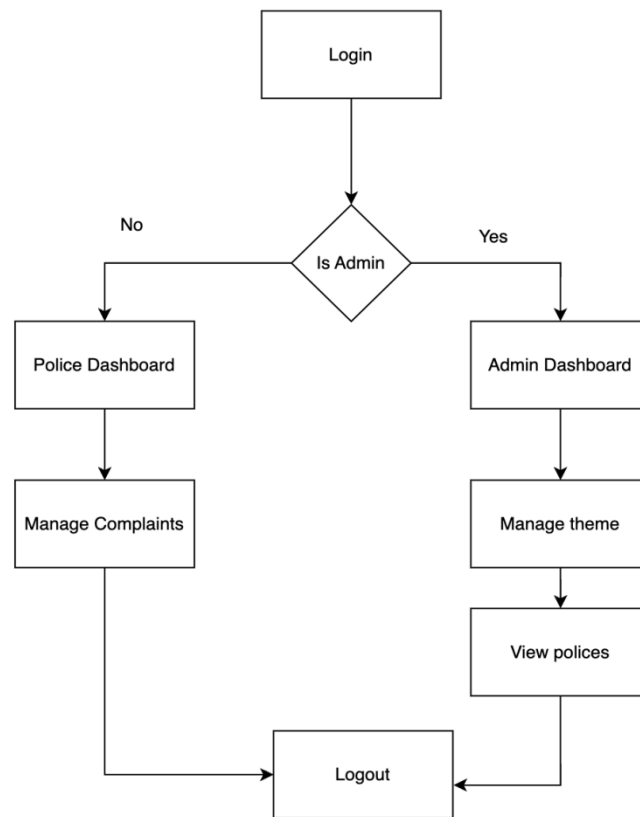


Figure 7 Activity of Crime Reporting System

3.2 System Design

The crime reporting system enables users to file complaints, while offering police stations to view complaints and take actions accordingly. The system architecture includes a user-friendly frontend, a robust backend for complaint processing, and a database for storing relevant information.

3.2.1 Deployment Diagram

A deployment diagram illustrates the physical architecture of a system, showing how software components (artifacts) are deployed across hardware nodes. It is primarily used in the design phase of system architecture to model the runtime structure of the system.

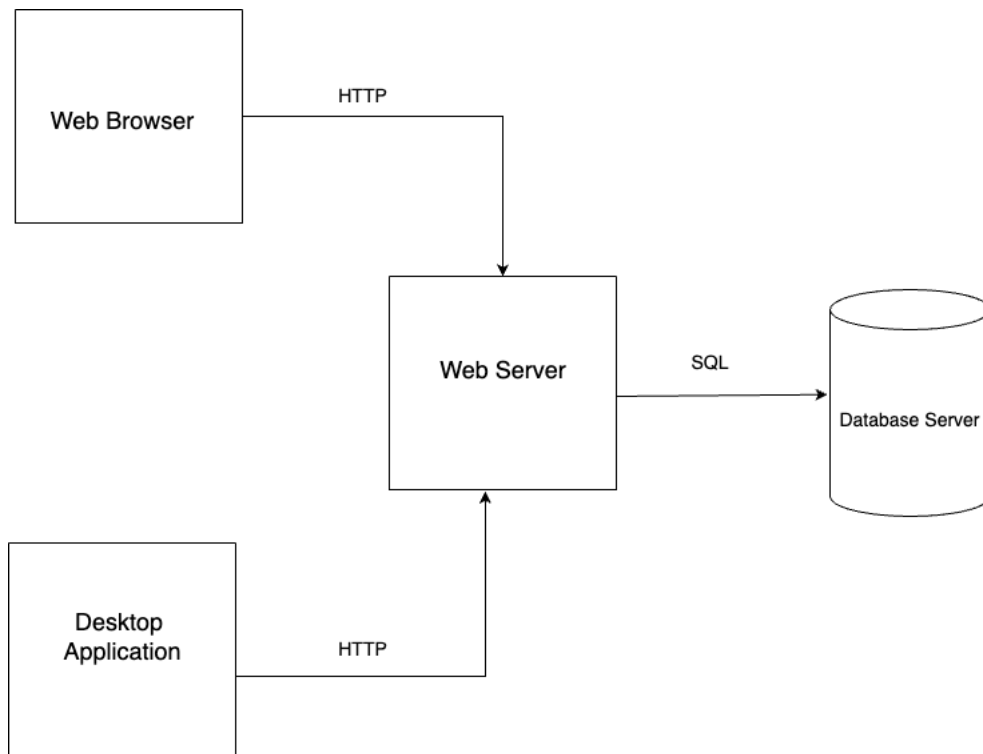


Figure 8 Deployment Diagram of CRS

3.2.2 Refinement of class and object

The figure below shows the refined class diagram for a Crime Reporting System. It includes three main classes: Police, Report, and Admin. Police can register, log in, browse complaints, and take actions. Admin manage their police stations and handle complaints , while admins oversee polices and complaints data. In the diagram, minus (–) indicates private access, and plus (+) indicates public access.

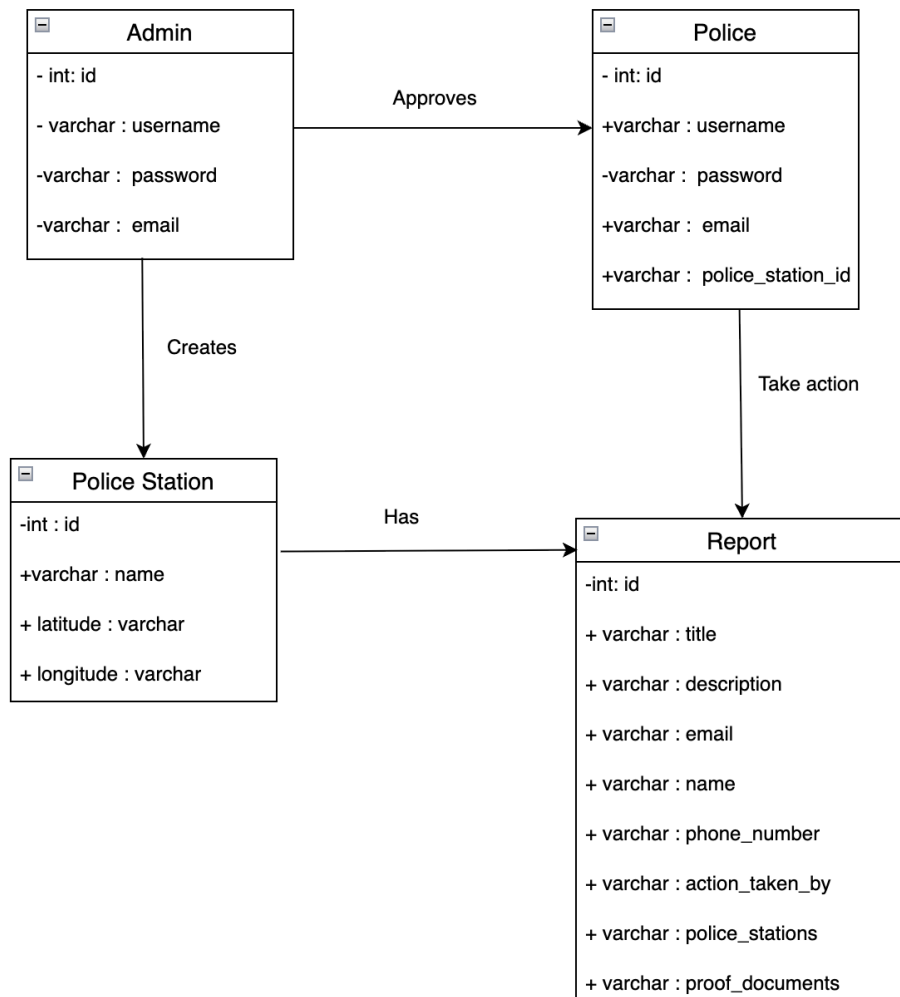


Figure 9 Refinement of Classes and Object Diagram for CRS

3.3 Algorithm Details

Haversine Algorithm

The algorithm used in my project is called the **Haversine Algorithm**.

The Haversine formula is a mathematical formula used to calculate the distance between two points on a sphere, given their latitudes and longitudes. It's particularly useful for finding the great-circle distance (shortest distance over the Earth's surface) between two points on Earth.

How It Works:

Comparing Police Station and Report Distance: The algorithm compares the distance between request user and police station and helps the user to send complain on nearest

police station.

Calculating Distance: It calculates the distance between 2 nodes . Police station and user.

Finding Closest Matches: The algorithm then ranks the police station based on their distance from the latitude and longitude. The police station with the shortest distance are the closest matches and are automatically choosed

Example:

If you are filling form from Kathmandu . Suppose there are many police stations. The police station which is nearest to the user is choosed automatically with this algorithm.

Formula :

$$a = \sin^2(\Delta\phi/2) + \cos \phi_1 \cdot \cos \phi_2 \cdot \sin^2(\Delta\lambda/2)$$

$$c = 2 * \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$d = R * c$$

Where:

- d is the distance between the two points (along the great circle).
- R is the radius of the sphere (e.g., the Earth's radius).
- ϕ is latitude, and λ is longitude, in radians.
- ϕ_1, λ_1 are the lat/long of the first point, and ϕ_2, λ_2 are the lat/long of the second point.
- $\Delta\phi$ is the difference in latitudes ($\phi_2 - \phi_1$).
- $\Delta\lambda$ is the difference in longitudes ($\lambda_2 - \lambda_1$)
- $\text{atan2}(y, x)$ is a function that computes the arctangent of y/x , taking into account the signs of both y and x to determine the correct quadrant.

Sorting Algorithm

Sorting complaints based on their status and date.

How it works:

First, we sort by the status (priority):

"action_taken" is the highest (priority 1), followed by "investigating" (priority 2), then "pending" (priority 3).

Then, if two complaints have the same status, we sort them by the date (newest first

Example:

If we have complaints like:

"action_taken" (most urgent)

"investigating"

"pending" (least urgent)

Complaints with "action_taken" will come first, followed by "investigatings," and then "pending." If two complaints are of the same status, the newest one will come first.

In short: Important complaints(like "action_taken") come first, and the newest ones show up first if they have the same status.

CHAPTER: 4

IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools used (CASE Tools, Programming Language, Database Platforms)

- **VS Code** VS Code is used as the primary Integrated Development Environment (IDE) for developing the crime reporting system. It provides powerful features like code editing, debugging which makes it ideal for developing the backend and frontend.
- **MS Office** MS Office is used for writing and editing the documentation of the crime reporting system. It helps in creating detailed reports, user manuals, and project documentation.
- **Django** : Django is a high-level Python web framework that enables rapid development of secure and maintainable websites. It follows the Model-View-Template (MVT) architecture and includes built-in features like user authentication, admin interface, and database integration, making web development faster and more efficient.
- **Draw.io** Draw.io is used to create diagrams for system analysis and design of the crime reporting system Flowcharts, data flow diagrams (DFD), and other design-related diagrams are made using this tool to visualize the system architecture and workflow.
- **Python** Python is a high-level, interpreted programming language known for its simplicity and readability. It supports multiple programming paradigms, including object-oriented, procedural, and functional programming. Python is widely used in web development, data analysis, machine learning, automation, and more due to its large ecosystem of libraries and community support.

4.1.2 Implementation Details of Module

Modules of this system are described below:

Admin Module The admin module is where the administrator can manage the entire crime reporting system. The features of the Admin Panel include managing polices, managing police stations, viewing complaints, updating the status of compaints, updating themes .

Admin Login

Admins log in with their email and password.

If the credentials are correct, they are redirected to the admin dashboard.

Dashboard

Displays an overview of the system: complaints, police stations, and users.

Links to pages for managing police, complaints, and police stations.

Manage Complaints

Admin can view the list of complaints, police stations.

Admin can add new police stations, edit existing items, or delete complaints and police stations.

Manage Users

Admin can view all registered police.

Admin can delete or update user information. Add and remove permissions on different modules.

Manage Complaints

Admin can view the list of all complaints, including users details such as user's name, email, and other personal details.

Admin can update the status of the complaints(e.g., from "pending" to "investigating").

User Module

The user module is where users can browse the public complaints. File a complaint and view its status.

Police Registration

New Police can sign up by providing their personal details (name, email, password, etc.).

The password is stored securely using hashing.

Police Login

Users log in with their username/email and password.

If successful, they are redirected to the police dashboard.

Dashboard Page

Displays a list of complaints fetched from the database.

Users can select the complaint and take action and later change the status and give explanations.

4.2 Testing

Unit Testing and System Testing are crucial steps in the software development lifecycle to ensure that each part of the system works as intended (unit testing) and that the system as

a whole functions correctly (system testing). Below are examples of Unit Testing and System Testing for a Crime Reporting System, focusing on both Admin, Police and User role.

4.2.1 Test case for Unit Testing

Unit Testing is a software testing technique in which individual components or units of a system—such as functions, methods, or classes—are tested in isolation from the rest of the application. The main objective is to ensure that each unit performs correctly according to its design and expected behavior. Unit tests are typically automated and executed frequently during the development process, allowing developers to detect bugs early, simplify debugging, and improve code quality. By testing units in isolation, developers can more easily pinpoint the source of errors, leading to faster issue resolution. Unit testing also supports better code design, encourages modular development, and provides greater confidence when refactoring code, as tests act as a safety net. Additionally, unit tests can serve as documentation by showing how individual units are expected to function. Popular tools for unit testing include JUnit for Java, PyTest for Python, NUnit for .NET, and Jest for JavaScript.

Admin Login

Table 2 Test case for Admin Login

Test name	Input	Expected output	Actual Ouput	Result
Invalid credentials	Username: admin, Password: admin123	Error message “invalid Credential”	Error message “invalid Credential”	pass
Valid credentials	Username: admin, Password: admin	Redirect to admin dashboard	Redirect to admin dashboard	pass

Admin Login Empty Username and password	Username: (empty), Password: (empty).	Error message: "Username and password cannot be empty"	Error message: "Username and password cannot be empty"	pass
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Police Login

Table 3 Test case for User Login

Test name	Input	Expected output	Actual Output	Result
Valid Login	user1, user123	Redirect to dashboard	Redirect to dashboard	pass
Invalid Username,password	Username:invalid, Password:invalid	"Invalid credentials"	"Invalid credentials"	pass
Empty username>Password	Username:(empty) Password:(empty)	"Username ,password cannot be empty"	"Username ,password cannot be empty"	pass

Police Registration

Table 4 Test case for user Registration

Test name	Input	Expected Output	Actual Output	Result
-----------	-------	-----------------	---------------	--------

User Registration - Valid	Username: user, Email: user@gmail.com , Password: user123, Confirm Password: user123	Success message: "Registration Successful"	Success message: "Registration Successful"	pass
User Registration - Invalid Email	Username: user, Email: user123, Password: user123	Error message: "Invalid email format"	Error message: "Invalid email format"	pass
User Registration - Password Mismatch	Username: user, Email: user@gmail.com , Password: user123, Confirm Password: user321	Error message: "Passwords do not match"	Error message: "Passwords do not match"	pass
User Registration - Empty Username,password,Email	Username,password, email: (empty)	Error message: "Username,password,email cannot be empty"	Error message: "Username,password,email cannot be empty"	pass

4.2.2 Test cases for System Testing

In system testing, whole System is tested as below:

Test plan

Table 5 Test case for Test Plan of Crime Reporting System

S.N	Test plan
-----	-----------

1	Check if the registration module works properly.
2	Check if the login module works properly.
3	To check if register crime module work properly.
4	To check if theme control module work properly.

Report Crime

Table 6 Test case for Crime Reporting System

Test name	Input	Expected output	Actual output	Result
File Complaint	Logged-in user, report a crime	Successfully reported a crime	Successfully reported a crime	pass
File Complaint (Invalid Data)	Invalid data some fields are missing	Redirect to same page	Redirected to same page	pass

Add Police Station

Table 7 Test case for Add police station of Crime Reporting System

Test name	Input	Expected output	Actual output	Result
Admin Adds police stations– Valid	name : Basundhara police chauki	Police station added successfully	Police station added successfully	pass
Admin Adds police station– Empty Field	name missing	Show validation error	No item added, error displayed	pass

CHAPTER: 5

CONCLUSION AND FUTURE RECOMMEDATIONS

5.1 Lesson Learnt/ Outcome

Working on the Crime Reporting Syste project has provided valuable experience and enhanced my knowledge in several key areas of software development. Throughout the project, I developed important skills such as effective problem-solving, independent research, communication, documentation writing, and time management.

- **Problem-Solving Skills** During the development process, I encountered various technical challenges and errors. This helped me improve my debugging and troubleshooting abilities while learning how to analyze issues and implement suitable solutions efficiently.
- **Writing and Documentation Skill** I gained experience in preparing technical documents such as project proposals and final reports. I also became familiar with using different CASE tools to design system models like use case diagrams, data flow diagrams, ER diagrams, class diagrams, and deployment diagrams.
- **Time Management** One of the key lessons from this project was learning how to allocate time effectively. I learned to prioritize complex modules (such as the complaint filing and tracking modules) and manage my schedule to ensure timely completion of development tasks.

- **Technical Skill Enhancement**

This project significantly improved my technical expertise, especially in web development technologies such as HTML, CSS, JavaScript, Python,Django, and PostgreSQL. I also gained hands-on experience working with databases and responsive UI design.

5.2 Conclusion

Information technology have changed the life of people, online medium is becoming the first choice in any commercial as well as non-commercial organizations. Website is becoming the inseparable part of the organization. CRS website can be valuable source of information and can be effective tool to improve communication, engagement, strength,

and build a solid trusting reputation within community. The interface as well as the backend functionalities of the system is user friendly and easy to use so any user with general knowledge of internet can easily use, manage and handle the system. The system doesn't need any complex tolls and hardware to run it can simply run on any internet browser available. The system will help managing records of crime reports from admin UI to Police as well . And admin can verify the report.

5.3 Future Recommendation

In the future, this Crime Reporting System can be enhanced with more advanced and user-friendly features to improve overall functionality and user experience. Some potential upgrades and additions include:

- **Live Meeting with Police Officers:** Integrate WEBRTC based real-time video calling feature with crime reporter and police officers.
- **OTP Verification for Complaints:** Add OTP confirmation for secure and verified complaints.
- **Advanced Reporting System:** Generate detailed reports of user complaints, police progress,police stations progress, and user feedback for analysis and improvement.

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APPENDICES I: SYSTEM SCREENSHOTS

The screenshot shows the CRS (Crime Reporting System) home page on the left and the 'Create Account' form on the right. The home page features the CRS logo, a quote by Albert Einstein, and a 'CRIME REPORTING SYSTEM' button. The 'Create Account' form is titled 'Create Account' and 'Register as a Police Officer'. It includes fields for First Name, Middle Name, Last Name, Email Address, Password, Confirm Password, and Police Station. A 'Create Account' button is at the bottom.

CRS
Crime Reporting System

Home Report Crime My Reports Police Login Police Signup Search reports...

Create Account
Register as a Police Officer

FIRST NAME
First name

MIDDLE NAME
Middle name

LAST NAME
Last name

EMAIL ADDRESS
Enter your email

PASSWORD
Enter your password

CONFIRM PASSWORD
Confirm your password

POLICE STATION
Select a station

Create Account

The screenshot shows the CRS (Crime Reporting System) home page on the left and the 'Report a Crime' form on the right. The home page features the CRS logo, a quote by William E. Gladstone, and a 'CRIME REPORTING SYSTEM' button. The 'Report a Crime' form is titled 'Report a Crime' and 'Fill in the details below to submit your report'. It includes fields for Name, Email, Phone Number, Incident Date & Time, Report Title, Description, and Proof Document. A 'Submit Report' button is at the bottom.

CRS
Crime Reporting System

Home Report Crime My Reports Police Login Police Signup Search reports...

Report a Crime
Fill in the details below to submit your report

NAME
Enter your full name

EMAIL
Enter your email address

PHONE NUMBER
Enter your phone number

INCIDENT DATE & TIME
dd/mm/yyyy, --:--

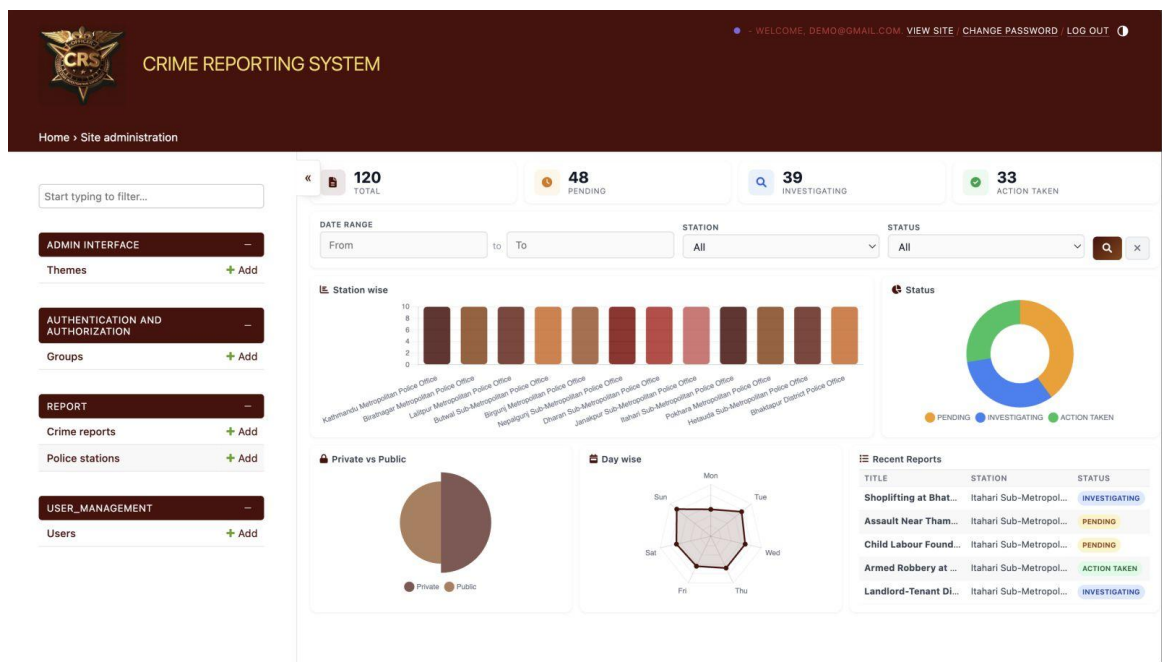
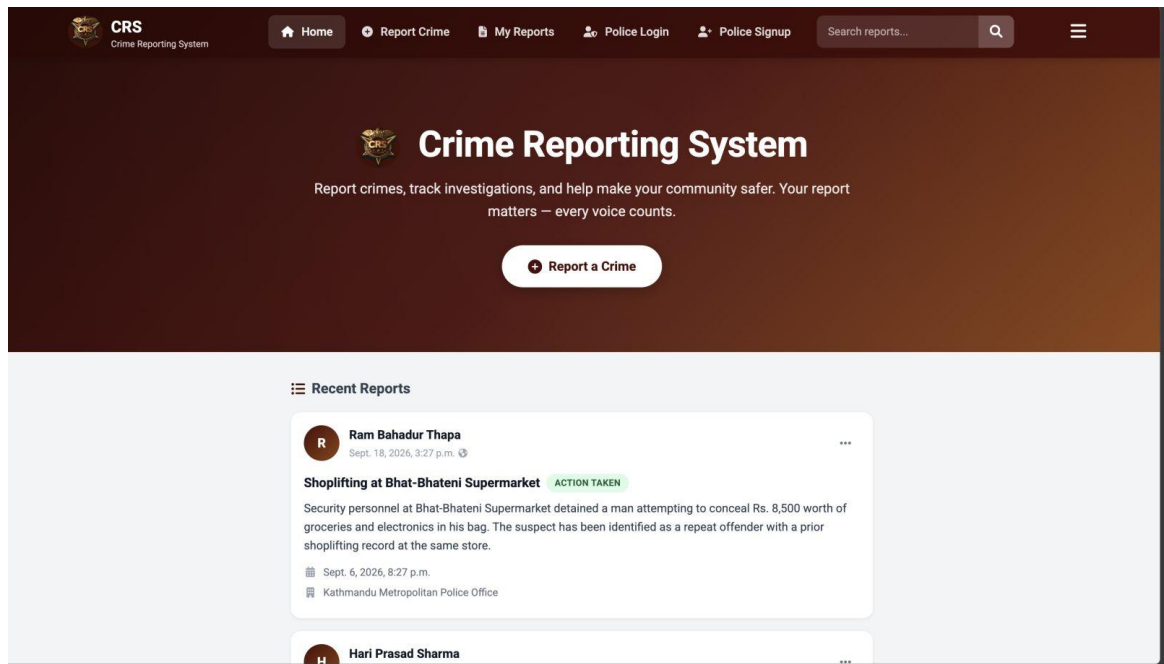
REPORT TITLE
e.g., Theft at Main Street

DESCRIPTION
Provide detailed information about the incident...

PROOF DOCUMENT
Choose File No file chosen

☒ Mark as private
Private reports will only be visible to you and administrators

Submit Report



Start typing to filter...

ADMIN INTERFACE

Themes

+

Add

AUTHENTICATION AND AUTHORIZATION

Groups

+

Add

REPORT

Crime reports

+

Add

Police stations

+

Add

USER_MANAGEMENT

Users

+

Add

Hetauda Sub-Metropolitan Police Office ---> Hetauda, Bagmati

HISTORY

Name: Hetauda Sub-Metropolitan Police Office

Location: Hetauda, Bagmati

Latitude: 27.4287

Longitude: 84.9962

Select Location

Search a place, click on the map, or drag the marker to set coordinates

Search location...

Lat: 27.4287 Lng: 84.9962

CRS

CRIME REPORTING SYSTEM

WELCOME, DEMO@GMAIL.COM VIEW SITE CHANGE PASSWORD LOG OUT

Home > Report > Crime reports

Start typing to filter...

ADMIN INTERFACE

Themes

+

Add

AUTHENTICATION AND AUTHORIZATION

Groups

+

Add

REPORT

Crime reports

+

Add

Police stations

+

Add

USER_MANAGEMENT

Users

+

Add

Select crime report to change

ADD CRIME REPORT

Q

Search

Action: -----

Go

0 of 100 selected

TITLE	PHONE NUMBER	CRIME DATETIME	STATUS	ACTION
Shoplifting at Bhat-Bhateni Supermarket	9841000020	July 20, 2026, 8:27 a.m.	Investigating	Edit
Assault Near Thamel Nightlife District	9841000019	July 20, 2026, 9:27 a.m.	Pending	Edit
Child Labour Found at Brick Kiln	9841000018	Sept. 11, 2026, 11:27 a.m.	Pending	Edit
Armed Robbery at Local Tea Shop	9841000017	Aug. 4, 2026, 9:27 a.m.	Action Taken	Edit
Landlord-Tenant Dispute Turning Violent	9841000016	July 30, 2026, 1:27 a.m.	Investigating	Edit
Drug Possession - Hashish	9841000015	Sept. 14, 2026, 7:27 a.m.	Pending	Edit
Burglary at Residential Property	9841000014	Aug. 8, 2026, 4:27 p.m.	Investigating	Edit
Fake Travel Agency Scam	9841000013	Sept. 15, 2026, 2:27 a.m.	Action Taken	Edit
Theft from Construction Site	9841000012	Aug. 30, 2026, 2:27 a.m.	Action Taken	Edit
Rash Driving and DUI	9841000011	July 25, 2026, 3:27 a.m.	Action Taken	Edit
Rash Driving and DUI	9841000010	Aug. 7, 2026, 8:27 a.m.	Pending	Edit
Fake Travel Agency Scam	9841000009	Aug. 20, 2026, 6:27 a.m.	Action Taken	Edit
Suicide Attempt Rescue	9841000008	Sept. 8, 2026, 1:27 a.m.	Investigating	Edit

FILTER

Show counts

By status

All

By crime datetime

Any date

By report taken by station

All

